

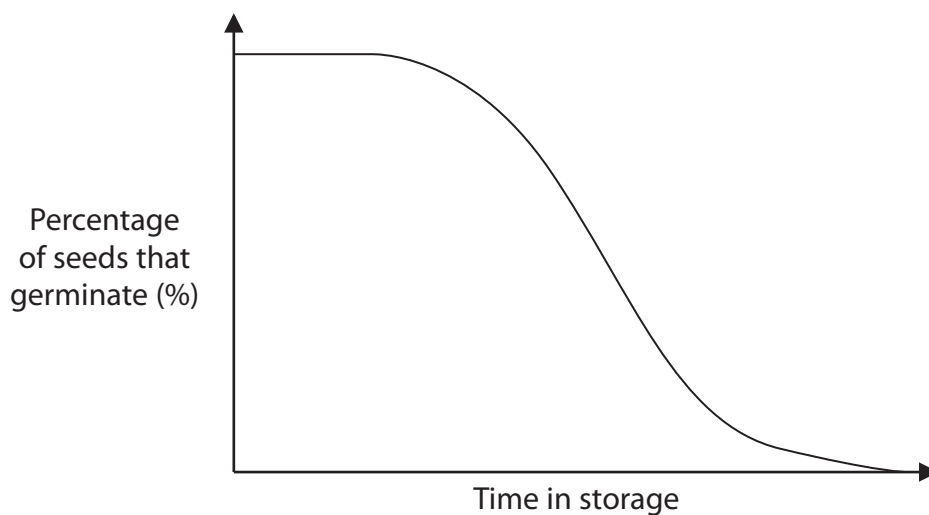
- 4 Seed banks are used to store seeds from a wide variety of plants. Some plants are a sustainable source of bioplastics.

(a) The Germplasm Bank of Wild Species is the largest seed bank for wild species of plants in Asia.

Seeds from 8 500 species have been collected and stored.

Seeds were dried before being stored at temperatures below 0 °C.

The effect of length of time in storage on seed germination is shown in the graph.



- (i) Describe the effect of length of time in storage on the germination of these seeds. (2)

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(ii) Two of the methods used in seed banks are:

- store seeds from many different plants of the same species, instead of many seeds from just one plant
- regularly germinate samples of the stored seeds, allowing them to grow into adult plants.

Explain the advantages of these methods.

(4)

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- (b) The photograph shows bioplastic straws and cutlery, produced from the seeds of avocados.



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- (i) In 2018, a company in Mexico produced 130 000 kg of bioplastic cutlery and straws per month.

40% of the products were straws.

Calculate the mass of cutlery produced per year by this company.

Give your answer to **two** significant figures.

(2)

Answer

- (ii) The use of these plant-based products is more sustainable than the use of cutlery and straws made from oil-based plastic.

Explain what is meant by the term sustainable, with reference to the cutlery produced from the seeds of avocados.

(2)

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(Total for Question 4 = 10 marks)

